



We've all experienced power failures at some time or the other. Whether you live in a metro that has relatively dependable power or in a smaller city with regular power outages, electricity problems can throw a spanner in the works of our gadget-dependent lives. In fact, electricity is the life force behind most of the work we do daily.

Office and home desktop computers, company networks and communication systems such as phones and fax systems—if any of these undergo a downtime during working hours, it can spell a substantial loss of money and, worse still, data. Besides, with the arrival of the monsoons, the reliability of power delivered to our homes and offices plummets to an all-time low.

Without Power...

Just as in any other combat situation, you need to know what you are up against before choosing your line of defence. There's a general misconception that a power failure involves just one situation—a blackout. This is but one of several problems that can be attributed to power failure. Here are the different types of power fluctuations that can occur:

■ **Blackout:** This is when there is a total power failure. It is akin to a light being switched off—the power from your electrical outputs falls from normal level to zero.

■ **Brownout:** This is a condition where the output voltage drops to a lower level than its normal value (230 Volts in India) for an extended period of time. This can affect the operation of electronic equipment and even

cause damage.

■ **Power surge:** Here, the power level rises above the normal value for a period and then drops to normal. This condition can be detrimental to electronic equipment—while most computer hardware can handle slightly higher input voltages over small periods of time, powerful surges can spell instant death for electronics if they are not protected.

■ **Voltage spike:** This is a condition where there is a brief impulse of a very high voltage, after which it returns to the normal. This condition can be lethal to sensitive electronic equipment, including computers. Such conditions often occur during rainstorms—when a bolt of lightning hits a power line, for instance.

■ **Frequency variation:** Here, the frequency of the AC voltage differs from its normal value of 60 Hz (the default frequency for conventional AC power in India). This can cause electronic equipment to malfunction. Computers are especially affected, as they are so dependent on voltage frequency.

Each of these faults in power can be caused either due to faulty wiring (lack of proper earth for your power lines), faults in the power lines themselves, errors in the power transmission grid and weather conditions such as lightning and storms.

...But Not Powerless

The implications and the extent to which each of these conditions can affect computer hardware are different. Accordingly, there are different solutions available to counter them.



Buying tips for UPS

A UPS is specified by its VA rating. This is a measure of the amount of power the UPS can provide—usually for about 10 to 12 minutes. Therefore, if a 500 VA UPS is loaded with equipment that draws 500 VA of power, it will last for about 10 minutes. If the load is halved to 250 VA, the backup time will increase to 16 to 18 minutes. This is assuming, of course, that the battery is fully charged and functional. Hence, when buying a UPS system, you should first ascertain what equipment is going to be connected to it and, more importantly, what kind of power conditions you want to protect your equipment from.

UPS systems should be used to power only the most critical parts of your computer. The following points need to be considered while buying a UPS:

■ **Input voltage range:** This is the range beyond which the UPS will deliver power through its batteries. The greater the range the UPS can handle, the more versatile its power protection. Look for a voltage range between 150 and 270 V.

■ **VA rating:** This is a rating of the power that can be provided by the UPS. You should choose the rating on the basis of the

requirements of the equipment you are going to connect to the UPS. As a rule of the thumb, a 1 GHz processor-based machine with a 20 GB hard disk, CD-ROM drive and 17-inch colour monitor would require a 500 VA UPS.

■ **Backup time:** This is the specified time for which the UPS can provide power from its battery for a given load. Look for a backup time of at least 10 minutes at maximum load.

■ **Switching/transfer time:** Used interchangeably, switching or transfer time is the time taken by the UPS to switch from normal delivery power mode to battery mode when it detects any power anomalies. Ideally, look for transfer times under 5 milliseconds.

■ **Alarms and connectivity:** Most UPS systems have LEDs on the front panel that indicate the state of its batteries and the condition of the input and output power. This visual indication of its operation is important, especially during power failures, so look for a UPS that has these indicators.

■ **Charging time:** This is the time taken to charge the UPS batteries from a completely drained state to a fully charged condition. This should be between 4 and 6 hours.